ModelIt!

Teacher's Guide

How Does Your Phone Hear Your Voice?

The Invisible Science of Sound Waves

Quick Reference

Grade Level	Grade 4
NGSS Standard	4-PS4-1
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: 4-PS4-1

Develop a model of waves to describe patterns in terms of amplitude and wavelength and that waves can cause objects to move.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a computational model showing how sound wave strength, vibration size, and volume level are connected in a wave system.
Disciplinary Core Idea	**PS4.A Wave Properties** — Waves, which are regular patterns of motion, can be made in water by disturbing the surface. Waves of the same type can differ in amplitude and wavelength. Waves can cause objects to move.
Crosscutting Concept	**Cause and Effect** — Students explore how changing the strength of a sound (cause) directly affects vibration size and volume (effects) in a predictable pattern.

CAST Testing Alignment

MULTIPLE CHOICE:

A student plucks a guitar string gently, then plucks it hard. The hard pluck makes a louder sound. Which best explains why the sound is louder?

CONSTRUCTED RESPONSE:

Using what you know about sound waves, explain how your voice travels from your mouth to a friend's ear across the room. Use the words 'vibration,' 'sound wave,' and 'amplitude' in your answer.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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> **For Your Confidence**
>
> This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Sound is all around us every moment of every day, yet we cannot see it. Understanding sound waves helps us explain how we communicate, how music works, and how modern technology like phones and speakers turn invisible vibrations into the sounds we love.

What Makes Sound?

Every sound begins with a vibration. When you speak, your vocal cords vibrate. When you clap, your hands vibrate. When a drum is hit, the drum skin vibrates. These vibrations push air molecules back and forth, creating a chain reaction of moving air particles that spreads outward in all directions — these traveling vibrations are sound waves.

How Sound Waves Travel

Sound waves are longitudinal waves — the air molecules compress together (compression) and spread apart (rarefaction) in the same direction the wave is traveling. Sound needs a medium (material) to travel through. It moves through air at about 343 meters per second, faster through water (1,480 m/s), and even faster through solids like steel (5,960 m/s). Sound cannot travel through a vacuum (empty space).

Amplitude and Volume

Amplitude is the height of a wave — how far the air molecules move from their resting position. Larger amplitude means the molecules move farther back and forth, carrying more energy, which our ears detect as louder sound. When you whisper, the amplitude is tiny. When you shout, the amplitude is large. Volume is measured in decibels (dB). A whisper is about 30 dB, normal talking is 60 dB, and a rock concert is 110 dB.

How Microphones Work

A microphone is a device that converts sound waves into electrical signals. Inside a basic microphone is a thin membrane called a diaphragm. When sound waves hit the diaphragm, it vibrates back and forth just like the original source. These vibrations are converted into electrical signals that match the pattern of the original sound wave. This is how phones, computers, and recording studios capture sound.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L — **Locate the System**
Students identify sound wave strength as an external component and vibration size and volume level as internal components that respond to the strength of the sound.

E — **Establish Relationships**
Students determine that sound wave strength positively affects vibration size, and vibration size positively affects volume level — a direct cause-and-effect chain.

V — **Visualize & Model**
Students build a model showing how stronger sounds create bigger vibrations that produce louder volume.

E — **Evaluate Outcomes**
Students run whisper, normal voice, and shout scenarios to observe how changing sound wave strength affects the entire system.

R — **Revise & Extend**
Students add distance from source or material type to explore why sounds get quieter over distance or sound different underwater.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Hey everyone — talk to your hand right now. Can you FEEL the vibrations? That is actually sound waves hitting your skin!"*

Watch For

Have students talk into their cupped hands to feel vibrations. Then ask: How does your phone feel these same vibrations?

Slide 2: Learning Objectives

3 min

Say: *"Today we are going to figure out how your phone can HEAR you — even though it does not have ears!"*

Watch For

Have students read objectives aloud. Pre-teach 'vibration' and 'amplitude' with a ruler vibrating off a desk edge.

Slide 3: Big Question

3 min

Say: *"You cannot see sound. You cannot touch it. But somehow your phone catches it perfectly. How?"*

Watch For

Have students turn and talk: What do you think is happening in the air between your mouth and the phone?

Slide 4: LEVER Framework

2 min

Say: *"We are going to build a model of sound to solve this mystery."*

Watch For

Briefly preview each LEVER step. Remind students that a model helps us understand things we cannot see.

Slide 5: Activity 1: Components

8 min

Say: *"What makes a sound loud or soft? What parts of this system can we identify?"*

Watch For

Guide sorting of three components. Explain that sound wave strength is external because the SPEAKER controls how hard they push air.

Slide 6: Activity 2: Connections

8 min

Say: *"When you shout, are the vibrations bigger or smaller? What about the volume?"*

Watch For

Help students draw positive arrows showing the chain: more strength leads to bigger vibrations leads to louder volume.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let us test it! We will start with a whisper and end with a shout."*

Watch For

Have students predict first, then run simulations for whisper, normal, and shout. Compare the wave diagrams.

Slide 8: Discoveries

5 min

Say: *"Now we know the secret — sound is made of real vibrations, and louder means BIGGER vibrations!"*

Watch For

Summarize discoveries. Demonstrate with a tuning fork in water to make vibrations visible.

Slide 9: STEM Challenge

12 min

Say: *"Can you build something that SHOWS sound waves? Let us make the invisible visible!"*

Watch For

Distribute materials (bowls, plastic wrap, rice, rubber bands). Groups build sound catchers and test with different volumes.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Sound Wave Strength
- **INTERNAL:** Vibration Size, Volume Level

Relationships

- **Sound Wave Strength to Vibration Size:** POSITIVE (+) — A stronger sound pushes air molecules harder, making them move farther back and forth. More energy in means bigger vibrations — shout and the waves are big, whisper and they are tiny.
- **Vibration Size to Volume Level:** POSITIVE (+) — Bigger vibrations carry more energy to your ear (or a microphone). Your ear detects bigger vibrations as louder sound. This is why amplitude and volume are directly connected.

Key Discoveries

- Sound is created by vibrations — every sound you hear started with something moving back and forth very fast
- Bigger vibrations (larger amplitude) make louder sounds, and smaller vibrations make softer sounds
- Sound waves need something to travel through — they cannot travel through empty space
- Your phone's microphone has a tiny part inside that vibrates when sound waves hit it, turning sound into electrical signals

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Sound can travel through empty space"	Sound MUST have a medium (air, water, or a solid) to travel through. Unlike light, sound cannot travel through a vacuum because there are no molecules to vibrate. This is why there is no sound in outer space — even if a star explodes!
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Louder sounds travel faster"	Loud and quiet sounds travel at the SAME speed through the same material. A whisper and a shout both travel through air at about 343 meters per second. Volume only changes the SIZE of the vibrations (amplitude), not the speed of the wave.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Sound only travels in one direction"	Sound waves spread out in ALL directions from the source, like ripples from a stone dropped in water. That is why you can hear someone talking even if you are behind them or around a corner. Sound waves expand as a sphere from the source.